AIR INDIA FLIGHT 171 CRASH REPORT

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1. Introduction and Accident Summary

On June 12, 2025, Air India Flight 171, a Boeing 787 Dreamliner, crashed shortly after takeoff from Sardar Vallabhbhai Patel International Airport in Ahmedabad, India. The aircraft was scheduled for a 9-hour flight to London Gatwick. Seconds after lifting off, the aircraft lost all engine power, struck a chimney and a student canteen, and exploded. The crash resulted in the deaths of 241 people, including both pilots, with only one passenger surviving the impact.

2. Aircraft Details

- **Model:** Boeing 787-8 Dreamliner
- **Flight Number:** AI171
- **Status:** Fully functional and within operational limits prior to takeoff
- **Fuel Load:** 54,000 kg for a 9-hour journey
- **Flight Route:** Ahmedabad (AMD) → London (LGW)

3. Ahmedabad Airport Details

- **Runway Used:** Runway 23
- **Runway Length:** 3,500 meters
- **Surroundings:** Dense residential neighborhoods and the Army Medical Corps compound located 1.7 km from the runway.

4. Weather and Environmental Condition

- **Temperature:** 36°C (Hot day)
- **Air Density:** Thin air due to high temperature, requiring higher takeoff speeds.
- **Visibility:** 6 km
- **Winds:** Light winds at 240° at 6 knots.

5.1 Flight and Crew Details

The flight was a regular commercial service connecting Gujarat to the UK diaspora. Boarding was completed at 1:10 p.m., and the aircraft pushed back at 1:18 p.m..

5.2 Captain Details

The captain was Sumit Sabarwal, 56 years old. He was highly experienced with more than 15,600 total flight hours. Over half of his flying experience (7,800+ hours) was specifically on the Boeing 787 Dreamliner.

5.3 First Officer Details

The First Officer was Clive Kunda, 32 years old. He had 3,400 total flight hours, with 1,100 hours on the Dreamliner. He came from an aviation family; his mother had flown for Air India for thirty years.

6. Storyline

It was a hot afternoon in Ahmedabad on June 12, 2025. Air India 171 was heavy with 230 passengers and full fuel for London. Because of the heat, the pilots knew they needed the full length of the runway to lift the heavy jet. The takeoff roll was normal until the plane left the ground. Suddenly, at just 55 feet in the air, both engines went silent. In the cockpit, confusion reigned as one pilot asked why the engines were turned off. The plane, having lost its thrust, could not clear the nearby buildings. It clipped a group of trees and a chimney before slamming into the BJ Medical College buildings.

7. Sequence of Events Leading to the Crash

- **1:37:33 p.m.:** Takeoff clearance issued.
- **1:37:37 p.m.:** Takeoff roll begins.
- **1:38:33 p.m.:** Aircraft reaches V1 speed (153 knots).
- **1:38:35 p.m.:** VR reached; nose wheel lifts.
- **1:38:42 p.m.:** Aircraft reaches 180 knots at 55 feet altitude.
- **Immediate Event:** Fuel cutoff switches for both Engine 1 and Engine 2 are moved to the "OFF" position.
- **1:38:47 p.m.:** Ram Air Turbine (RAT) deploys to provide emergency power.
- **1:39:09 p.m.:** First contact with trees and the Army incinerator chimney.
- **1:39:10 p.m.:** Fuselage strikes Building A; wings shear off against Building B and C.

8. Swiss Cheese Model Analysis

8.1 Human Factors

- **Physical Interaction:** The manual movement of the fuel cutoff switches to the "OFF" position.
- **Communication Breakdown:** Cockpit voice recordings suggest one pilot was unaware of the other's actions regarding the switches.
- **Lack of Gear Retraction:** No "Gear Up" call was made because the pilots had no time to verify a positive rate of climb.

8.2 Technical Factors

- **Switch Design:** The fuel switches are mechanical; there is no software interlock to prevent them from being flipped during takeoff.
- **ELT Failure:** The Emergency Locator Transmitter (ELT) failed to activate, hindering immediate rescue identification.
- **Black Box Damage:** One flight recorder was destroyed by thermal damage, leaving only one source of data.

8.3 Environmental Factors

- **High Temperature:** 36°C heat made the aircraft's climb performance sluggish due to thin air.
- **Low Altitude Emergency:** The engine failure occurred at only 55 feet, leaving the crew with zero time to troubleshoot or restart.

8.4 Organizational Factors

- **Equipment Deficit:** The national investigation bureau (AIB) lacked the "golden chassis" cables needed to read the Boeing 787 data, causing delays.

9. Investigation Theories and Observations

- **Software Malfunction (TCMA):** This theory was **eliminated**. The TCMA software cannot physically move cockpit switches and would have prevented the engine restart seen in the data.
- **Accidental Activation:** Considered **unlikely**. The switches require a "pinch and lift" motion against spring pressure, and moving both within 1 second suggests a deliberate act rather than an accident.
- **Intentional Act: Confirmed**. Direct mechanical wiring recorded the physical movement of the switches to cutoff, meaning someone in the cockpit physically moved them.

10. Post-Accident Recommendations and Actions

- **Safety Lockouts:** Recommended that Boeing evaluate a software or mechanical "lockout" for fuel switches during the takeoff phase.
- **Equipment Standards:** A mandate for the AIB to maintain modern "golden chassis" hardware for all aircraft types operating in the country.
- **ELT Reliability:** Investigation into why crash-rated devices like the ELT and rear black box failed in a relatively low-speed impact.

END of Report